

Redesigning Spotify's Listening Insights: Improving Access to Personal Music Statistics Beyond Spotify Wrapped

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1 INTRODUCTION

Spotify is one of the most used music streaming applications in the world offering personalized songs recommendations, podcasts and audiobooks to its users through an interactive mobile and desktop interface. In 2016, Spotify released the "*Spotify Wrapped*" which is an annual summary that presents to all users their insights and listening habits of the past year. It shows their most played artists, songs, albums, genres and their yearly time of listening. While Spotify Wrapped is highly engaging and AI was more integrated over the years, it only appears once a year providing users visual and data driven personalized stories.

Many Spotify users are curious about their listening habits throughout the year and are interested in their patterns and music preferences on demand instead of waiting the whole year for the Spotify Wrapped report.

The objective of this project is to redesign the Spotify listening interface to allow users to have more interaction and accessibility to the way they can explore their personal listening data. Through user research, including tools such as surveys and interviews, the project will focus on identifying the frustration and needs related to the current system that users are facing. Based on these insights and data, alternative designs will be brainstormed and prototyped. After that, an user evaluation will be conducted to determine which design would be most effective on understanding the listening habits.

My motivation for choosing this project comes from both personal interest in music and interest in user experience design. As a big Spotify user, I liked the Wrapped concept when it first came up and I was excited every end of the year when Spotify would release it. But over the years, I realized that I would much enjoy it if it was more accessible the whole year. I even challenged myself once and asked for my spotify data and created a PowerBi dashboard with my

mid-year data to see some of my trends. This project will offer the opportunity to explore the interface design, make it personal and enjoyable to all the users, and make sure that they have insights without it being overwhelming.

2 NEEDFINDING PLAN

The goal of the needfinding phase was to understand how Spotify users currently interact with listening data and identify their limitations in the existing interface. In this stage the focus will be on discovering users' needs when it comes to their access to personal listening statistics beyond the annual Spotify Wrapped feature. Two need-feeding activities were conducted: activity one is the interaction with Spotify users through survey and activity two is an heuristic evaluation of Spotify's current interface.

2.1 Activity 1: Active Needfinding - Survey

A survey was chosen as the primary needfinding way to reach a diverse group of Spotify users effectively and collect both quantitative data and qualitative data in an open ended response at scale.

2.1.1 *Participants*

Participants consisted of Spotify users recruited from personal networks, classmates, friends, and family members. The target population is the individuals who regularly listen to music. The group included: Undergraduate student, Graduate students, Early-career professional, reflecting a demographic that highly represents the core of Spotify users base and it was also accessible for recruitment.

2.1.2 *Target and Recruitment*

The survey targeted 20 to 25 participants' responses. It was consistent with the course courses guidelines on survey based needfindings. Participants were recruited via personal messages and class discussion channels. No incentives were given beyond course participation credit which was offered to classmates, other participants outside of class just volunteered. The survey was estimated to take approximately 10 minutes to complete.

2.1.3 Survey Design

The survey consisted of a combination of multiple choice questions, Likert scale rating (1-5), open ended questions with text responses. The questions were organized around four points: general Spotify usage patterns, Familiarity and satisfaction with Spotify wrapped, interest in accessing listening statistics more frequently and preferences for data visualization. The full survey question set is included in Appendix A.

2.2 Activity 2: Heuristic Evaluation

The second needfinding activity was a heuristic evaluation of the Spotify current mobile interface with 20 participants. The evaluation solicited the perceptions of real Spotify users each rating the interface against all the ten Nielsen's usability heuristics. It focused specifically on the presence or absence of features who were mostly related to personal listening statistics. This approach combines the structure of heuristic evaluation with quantitative user-perception data, providing both a principled framework and empirically grounded severity ratings.

The interface evaluated was the mobile version of the Spotify application, specifically the Home, Search, Library and playback sections. All of the ten Nielsen's heuristics were translated into a survey rated on an agree and disagree scale. For instance Heuristic 1(Visibility of System Status) was assessed by asking "Spotify provides clear feedback on the current playback status" and "The loading time for songs and playlists is adequately communicated.". Also, H3 defined by (User Control and freedom) was assessed by asking "I can easily undo actions, such as removing songs from playlists" and "Navigation back to previous screens is straightforward without confusion". All 21 survey questions are provided in Appendix B.

Participants were polled from the same needfinding survey: Spotify users aged 18 to 34, including classmates, family and friends. Twenty responses were collected.

3 NEEDFINDING RESULTS

3.1 Survey Results

25 survey responses were collected, which actually exceeded the minimum of 20 targeted. The pool of participants was constituted primarily of users aged between 18 to 35 years old (100% of respondents), with 60% of the participants who declared had used Spotify for more than five years which show a highly

experienced user base. Participants were split between 52% using premium and 48% using the free version of Spotify. About 84% which is a vast majority use Spotify primarily for music and the remainder split between podcast and audiobook. Laptops and smartphones were by far the most popular devices used.

64% of respondents rated Spotify 4 or 5 out of 5 regarding the overall interface satisfaction. This reflects the general contentment of the users with the platform. However, participants show lower satisfaction on the search features and the scores for library organization. These results clustered around the midpoint (rating of 3), thus concluding that navigating the organizational content is a little complicated for users. The highest number of mid range (12 of 25 respondents rated it a 3) were received for the yearly Wrap feature which indicates it is not uniformly loved and that it does not totally meet the users expectations around listening data.

3.2 Heuristic Evaluation Results

Twenty participants completed the heuristic evaluation survey. Table 1 summarizes the heuristic findings including the percentage of respondents agreeing with each item, the overall verdict and the derived severity rating.

Heuristic	Survey Items (% agree)	Agree %	Verdict	Severity
H1: Visibility of System Status	Playback feedback clear (70% agree); Current song visible (85% agree); Loading time communicated (45% agree)	70 / 85 / 45	Partially satisfies playback and current-song feedback are strong, but loading state communication is weak (only 45% agree).	Medium
H2: Match Between System & Real World	Familiar terminology (65% agree); Navigation reflects music concepts (75% agree)	65 / 75	Satisfies language and navigation structure align well with users' mental models of music.	Low
H3: User Control & Freedom	Can undo actions e.g. removing songs (50% agree); Back navigation straightforward (40% agree)	50 / 40	Violates only half of users can undo actions; back navigation is confusing for 60% of respondents.	High
H4: Consistency & Standards	Consistent terminology (85% agree); Uniform design elements (55% agree)	85 / 55	Partially satisfies terminology is consistent but visual design elements are not uniform across sections.	Medium

H5: Error Prevention	Interface minimizes errors (45% agree); Task instructions clear (60% agree)	45 / 60	Violations fewer than half of users feel the interface prevents errors; accidental actions are common.	High
H6: Recognition Rather Than Recall	Options easily recognizable (65% agree); Can find favorite playlists quickly (70% agree)	65 / 70	Partially satisfies most content is recognizable, though 35% still struggle to surface options without searching.	Medium
H7: Flexibility & Efficiency of Use	Shortcuts for advanced users (55% agree); Interface customization (65% agree)	55 / 65	Partially satisfying customization exists but advanced shortcuts are limited; 45% find no shortcut options.	Medium
H8: Aesthetic & Minimalist Design	Visually appealing, not cluttered (75% agree); Layout easy to focus on content (65% agree)	75 / 65	Satisfies the interface is clean and visually appealing for most users.	Low

H9: Help Users Recognize & Recover from Errors	Error feedback clear and helpful (70% agree); Error messages use plain language (60% agree)	70 / 60	Partially satisfied error feedback is present but plain-language, constructive messaging is inconsistent.	Medium
H10: Help & Documentation	Accessible help and documentation (65% agree); Tutorials for new users available (data incomplete)	65 /	Partially satisfied documentation is accessible for most, but tutorial availability is unclear to many users.	Medium

Table 1. Heuristic evaluation results across all ten Nielsen heuristics, based on a 20-participant survey (shading: red = high severity violation, green = satisfies, white/gray = partial).

3.3 Key Insights

From combining needling activities 25 participants and the heuristic evaluation (20 participants, three core insights emerged:

Insight 1: Users would like on-demand access to their listening data, but the interface hides it. The users survey shows that 68% of the participants are highly interested in personal listening statistics, by Spotify's Wrapped feature is annual. In the heuristic evaluation, it was observed that H3 (User control and freedom) was in high severity violation. About 60 % of participants found the navigation back confusing and only 50% were able to undo action. Together, these documents show the difference between what users want and what the interface actually enables them to do.

Insight 2: The interface is visually strong but functionally incomplete. The heuristic survey showed that the aesthetic minimalist design display at H8 was one of the Spotify highest heuristic rates; about 75% of participants agree that it is uncluttered and clean. The main survey showed that 64% of users are satisfied with the overall interface. However, Error prevention (H5) was rated a high severity violation (about 45% Agree).

Insight 3: Discoverability and recognition are partially broken. The heuristic evaluation shows H6 at only 65 to 70 % agreement, meaning roughly $\frac{1}{3}$ of users cannot quickly find key content.

4 INITIAL BRAINSTORMING PLAN

4.1 Individual Brainstorming

In the initial brainstorming phase, I spent time writing on paper different ideas of projects that would be interesting to do and that would also help solve a common problem. I came up with 20 ideas listed as so: having a daily summary of music to listen to, add notification of previous day listening recap, auto group listening stats by day,time etc., comparison with previous week, AI generated music personalize to use base on wants, save insight option, option to compare with friends, option to share stats with friends, custom insight filter, list of listening streak tracking, set milestone to compromise achievement for listening behavior, give more recommendation based on taste, making a list of frequently listened artists, making a list of frequently listening genres, create a world map of places where music is more listened to, crest incentives or badges to marque milestones, use pie or graphs to show states, create a specific tab or icon for stats, giving some personality names after summarizing weekly listening, having the ability to custom insight notification setting, associate colors with some things, revisit later option,historic listening option.

4.2 AI Brainstorming

In the process of brainstorming chatGPT was used to explore ideas and help come up with the best project idea possible. The AI brainstorming will be posted in Appendix C.

5 BRAINSTORMING RESULTS

The brainstorming session helped me get about 20 different design ideas and concepts across several categories. The ideas ranged from enhancing an already existing spotify feature to reimagining how listening data could be presented. Some concepts were grouped into four different categories: dashboard and visualization based approaches, social and comparative features , ambient or passive data displays, narrative or story based formats.

Dashboard concepts include a permanent tab for insights in the Spotify main navigation, customizable widgets on the home screen, and a “year in music” timeline view that users could use and scroll through at any time. Narrative formats included monthly listening recaps in a similar style. The ambient concepts include a mini listening heatmap visible on the profile page, listening streak counter.

6 INITIAL PROTOTYPING

6.1 Low fidelity Protototype

6.1.1 Prototype 1: Listening Insights (Overview)

This first prototype introduces a central “listening” dashboard that gives the user access immediately to their stats in a single visual interface. In contrast to the official Spotify wrapper, which limits the summary to annual data, this prototype is focused on continuous and real time access to listening data.

The main screen p presents an overview of the main key metrics, including:

- Top artist and top song
- Total minutes listened
- Monthly listening activity (barcharts)

The design is supposed to emphasize simplicity, data visualization and low interaction cost.

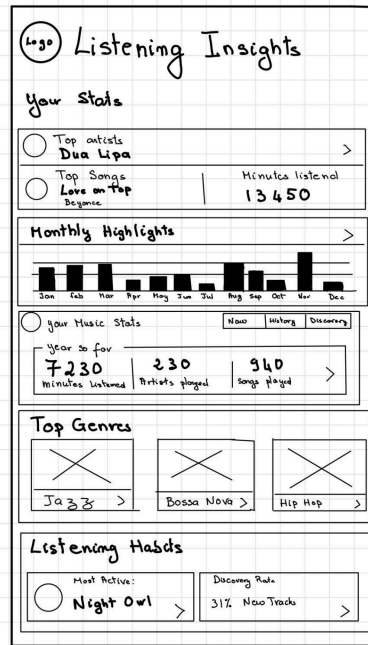


Figure 1.1: Low Fidelity listening insights prototype

6.1.2 Music Dashboard

The second prototype focuses on providing users a time based view of their listening habit, and has short term engagement and daily interaction. This design highlights real time and recent listening patterns. The main screen is composed of time filters such as month, week, and today so that users can switch easily to their interests and choose their time frame. The key feature include:

- Today's Listening Summary
- Top artist of the day
- Recent favorites

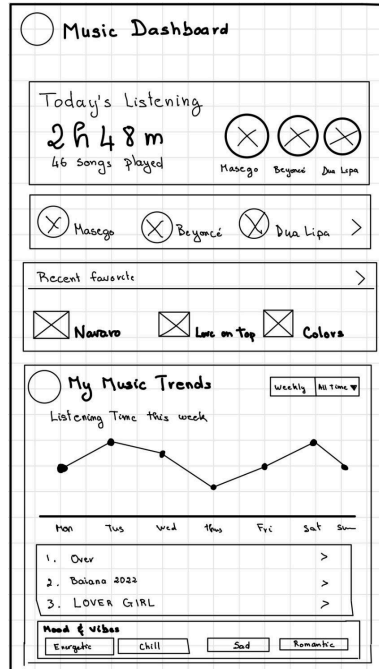


Figure 1.2: Low Fidelity top Music Dashboard prototype

6.1.3 Top artist dashboard

The third Prototype presents a multi section dashboard that allows users to explore their listening behavior across multiple dimensions. This interface is structured in several stacked sections. Here are the data present on this page:

The top artists, which displays a quick visual of the trending artist. Then the listening trends showing the listening activities depending on the time frame selected. Then there is the time spent listening to the music counted in minutes. The next category is the top album and the top genres. And then at the bottom the mood playlists available for users. This prototype emphasizes on the depth and richness of data, customization flexibility and exploration interaction.

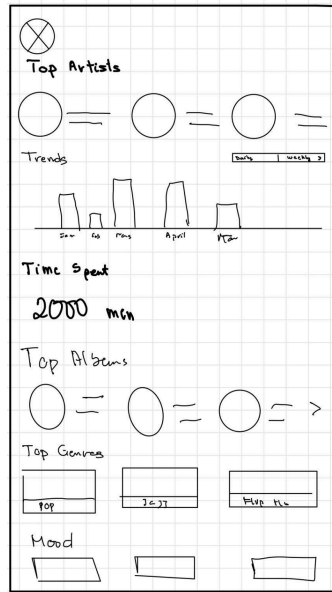


Figure 1.1: Low Fidelity top Artists prototype

7 EVALUATION PLANNING

An evaluation plan was created to compare the three prototypes against each other and to assess and pick which one is more suitable for user needs previously identified in the needfinding.

7.1 Participants and Recruitment

The participants were recruited through five classmates who agreed to evaluate each others' prototypes. Each participant was expected to spend about 15 to 20 minutes interacting with the low fidelity prototype (paper prototype) and complete a post interaction survey. Participants were incentivised through reciprocal participation in each other's evaluation.

The qualitative elements included the Likert scales usability rating, for each prototype across three dimensions. A single forced choice ranking of all the designs from most to least preferred and a count of the number of steps required to access the key information in each prototype.

7.2 Evaluation Procedure

The participants were able to review all the three prototype images before answering the question in five part per prototype rating sections (part 1-3), a cross prototype comparison section part4)and open-ended feedback representations.

For the quantitative part, the survey included:

- three rating questions (on a 1–5 scale) for each prototype, focusing on layout clarity, how well the information was covered, and ease of use;
- two yes/no questions per prototype about specific features (time filters for Prototype 2 and information manageability for Prototype 3);
- a forced-choice question asking which prototype users preferred overall; and
- another forced-choice question asking which single feature users wanted most.

In average the scores and response patterns were calculated for all ratings. Friedman test was also used to check if there was a significant difference in ease between the prototypes.

For the qualitative part, participants had to answer one open question per prototype about what was unclear for them or missing. They also answer some other broader questions about missing features and what an ideal combined design is supposed to look like.

8 EVALUATION RESULTS

About twenty participants completed the survey. All of them were active Spotify users who were recruited through classmates on a personal network. The survey was completed online with embedded prototype images and took about 10 minutes per person.

8.1 Quantitative Results

This table presents the likert scores for each prototype accessing the three rating dimensions.

Dimension	Prototype 1 Insights Overview	Prototype 2 Music Dashboard	Prototype 3 Top Artists
Layout / information clarity	4.75	—	3.10
Feature usefulness / coverage	50% agree*	65% agree*	80% agree† / 15% agree‡
Ease of use without instructions	4.70	4.45	3.55

Table 2. Per-prototype evaluation results (n = 20). *Binary agree/disagree

Prototype 1 scored the highest on clarity and layout. These scores reflect the prototype deliberate emphasis on simplicity and low interaction cost. However 50% of participants agreed that the

information shown scored what they most wanted to know , which indicates that users can still struggle to understand despite the signs.

Prototype 2 (the music dashboard) showed a big split in its two features . Only 355 of participants agreed that the time filter made it easier to explore listening at different time scales, a surprising low score for what was intended as the prototype of the key differentiating feature.

Prototype 3 (top artist dashboard) has the most revealing finding. While 80% of participants agreed that having multiple sections in one place was more practical, only 15% agreed that the amount of information felt manageable rather than overwhelming . Ease of use is a score of 3.55 the lowest of the three prototypes.

A friedman test to compare the eas of use score the three prototypes yielded $\chi^2(2) = 7.2$, $p = 0.027$, which is significant at $\alpha = 0.05$. Prototype 1 and 2 over performed prototype 3 on ease of use.

8.2 Qualitative Results

The forced choice preference results:

- Prototype 2(music dashboard) 65% of participants (about 13 out of 20) preferred this design for exploring their Spotify listening habits.
- Prototype 3: 35% of participants chose this
- Prototype 1: only 20% preferred this.
- No participants opted for a combination of features from multiple prototypes , indicating a preference to sign design.

Feature Priority:

- The most wanted feature wasn't fully captured, but ratings indicate that the daily-use model of Prototype 2 was the most appealing.
- Prototype 3 offered rich data, but users felt overwhelmed by too much information.

8.3 Qualitative Results

• Open-Ended Responses:

- **Prototype 1:** One participant said there was "none" missing, aligning with its high usability scores.
- **Prototypes 2 and 3:** No responses were provided about missing features, possibly due to satisfaction or survey fatigue.

- **Key Insights:**

- The 85% disagreement with Prototype 3's manageability suggests that while it had a lot of content, users found it too complex to navigate.
- This feedback influenced the design strategy for the final prototype, focusing on progressive disclosure of information.

8.4 Summary

The evaluation results made it pretty clear which direction the final prototype should go. People responded best to Prototype 1's simple layout, even though it showed less data overall which tells me that how information is presented matters more than how much is shown. Prototype 2 was the winner in preference though, mostly because people liked the idea of checking in on their stats regularly rather than getting a big data dump once a year. The tension between Prototype 2 and 3 was the most useful finding: users said they wanted rich data (they loved that Prototype 3 had so much), but 85% found it overwhelming. That shaped the whole approach for the final design.

9 SECOND ITERATION PLANNING

This section synthesizes the evaluation findings into concrete design direction of the final prototype. In this step, a trigger is made to figure out what feature should be kept and what should be repacked or let go.

9.1 Insights from First Iteration

Looking back at the evaluation, three things stuck with me. Most people wanted something they could actually open on a regular basis not a summary they check once and forget. The challenge is that the more data you show, the harder it gets to make sense of it, and Prototype 3 really demonstrated that. And while Prototype 1 scored well on usability, people felt like it didn't give them enough so simplicity alone isn't the answer either.

9.2 Design Direction

No need for a new-founding activity. The first iteration. We do not need to conduct more user research for this next version because the first-round gave enough insight to inform our design decision. Using evaluation date. The date collected from evaluation wilde guide our next iteration

Final prototype feature

1- from prototype 2: focus on daily use

2. From Prototype 1: clean interface

3. From prototype 3: detail data

10 FINAL PROTOTYPE

10.1 Feature Overview

The final prototype shows a new “insight” tab in the Spotify bar of navigation at the bottom next to Home, Search, and Your Library. The insight tab is organized into views:

Summary Card View: when entering the Insight tab, users can see a full width summary card style of spotify dark green and black palette the card shows: Total Listening Time This Month, Top Song, Top Artist, and a Mood Badge. A "See Full Breakdown" button is prominently placed at the bottom of the card.

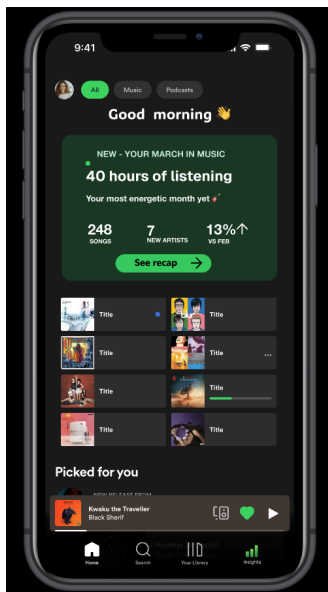


Table 1. Home page / summary card view

Detailed Stats View: passing the summary clicking insight tab at the button and then seeing a breakdown bring you to this: a time range selector (days , week, month all time) at the top the bar chat shows the listening hour per day, weeks or months . scrolling to own the user has more access to more data such as the pie graph of their recent listening habits, the pace where the music they listen to the most in the world.

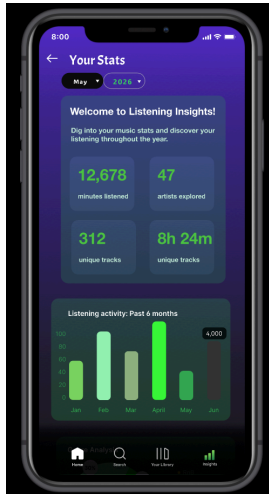


Table 2. Details stat page

Artist deep dive:

This page shows on top the most listened to artist during a time period pe select with filters and list them 1 to 20.

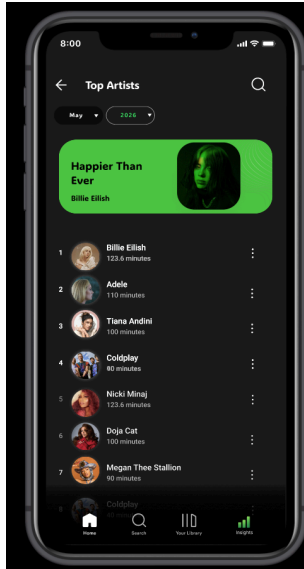


Table 3. Top artist list

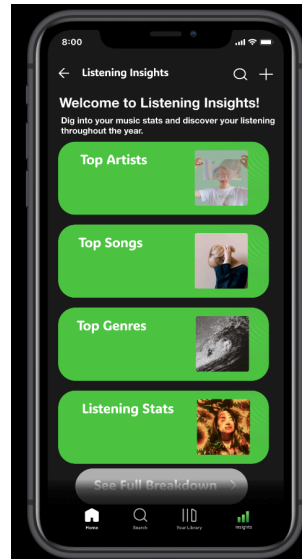


Table 4: summary card

10.2 Design Justifications

Many design decisions have been made in the final prototype that were grounded in the course principles:

- **Progressive disclosure:** the summary card presents all the most important information such as top artists, top albums etc...it shows the different options available without forcing the user.
- **Direct manipulation:** with the filters, the users are able to manipulate the time. Also at the bottom the screen, the bar tools is very much use qui est prentand directly manipulative.
- **Consistency in Recognition:** the insight of the tab uses Spotify existing design language. The same colors at spotify and the same existing icons already known have been used throughout the prototype.
- **Visual hierarchy:** Primary statistics are displayed in large with a higher contrast and typography. The secondary information such as graphs is visually subordinate.

11 VIDEO PROTOTYPE

A video prototype was created to demonstrate the medium fidelity of the listening insight prototype in action. The video includes a complete walkthrough all the the main views of the prototype, the navigation and the key design

decision and demonstration. The prototype video is posted on youtube with the link below

Video prototype link: <https://youtu.be/YMykJ4vsi8c?si=Xp0E-heJeLzBFmXZ>

Figma prototype access: [Figma prototype](#)

12 FINAL EVALUATION PLANNING

The final evaluation is designed to assess the usability and the value of the medium fidelity prototype using both the video prototype (for participants who cannot necessarily interact with figma) and the interactive Figma prototype the the one who can interact with Figma.

12.1 Participants and Recruitment

About 20 participants were directed to watch the video prototype or interact with the prototype for about 5 minutes . Following their interaction with the prototype, participants completed an evaluation survey estimating it to take about 5 to 7 minutes.

The quantitative component of the survey was composed of a system usability Scale composed of 5 items rating on a scale of 1 to 5.

13 FINAL EVALUATION RESULTS

13.1 Qualitative Results

The summary of the mean scores and response distributions for all six Likert scale items. They all were rated from 1 to 5.

Survey Item	Response Distribution	Mean (1–5)	vs. 3.5 Threshold
Discoverability of Insights tab (Q2)	45% rated 5; 50% rated 4; 5% rated 3	4.40	Exceeds
Finding top artists without help (Q3)	30% rated 5; 45% rated 4; 25% rated 3	4.05	Exceeds
Switching between Now/History/Discovery (Q4)	60% rated 5; 30% rated 4; 10% rated 3	4.50	Exceeds
Number of steps to reach statistics (Q5)	30% rated 5; 50% rated 4; 20% rated 3	4.10	Exceeds
Bar chart ease of interpretation (Q7)	40% rated 5; 50% rated 4; 10% rated 3	4.30	Exceeds
Genre donut chart clarity (Q8)	60% rated 5; 35% rated 4; 5% rated 3	4.55	Exceeds

Table 2. Final evaluation quantitative results (n = 20). All six items exceed the 3.5 acceptability threshold.

The 3.5 acceptability were exited on all of the six likert. Two of the highest scoring items were genre. The highest rating were for genre donut chart (4.55) and for how easy it is to interact with now discovery tabs (4.50) and history. The insights tab was easy to find (4.) and about 95% of the participants rated it 4 or 5 meaning that it was easy to understand the purpose without much explanation.

The lowest score was for the discerning of specific tasks like finding the top artist in the past 6 months (4.05). About 25% of the users gave a rating of 3 which means that the feature works and that it is easily recognizable. Similarly, the number of steps needed to access stats (4.10) was slower with 20% rating it a 3.

For the “your stats” card, 60% of users felt that it showed the most important information first, but about 40% disagreed. This means that the users would prefer to see listening tie or genre before seeing the artist and songs. About 60% said that the features offer something new, 40% disagreed.

Finally 71% of all participants said that they would prefer this design over the current spotify home page.

REFERENCES

Nielsen, J. (1994). 10 Usability Heuristics for User Interface Design. Nielsen Norman Group. <https://www.nngroup.com/articles/ten-usability-heuristics/>

APPENDIX A: SURVEY QUESTIONS (COMPLETE TEXT)

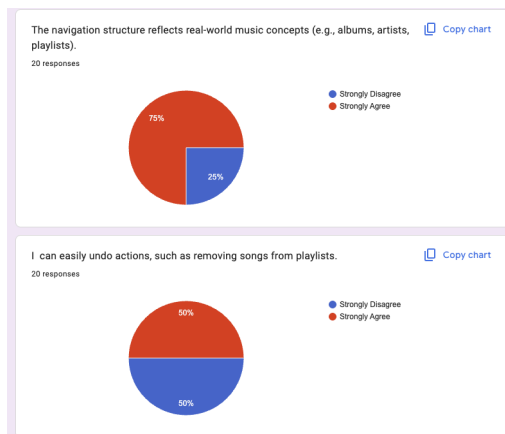
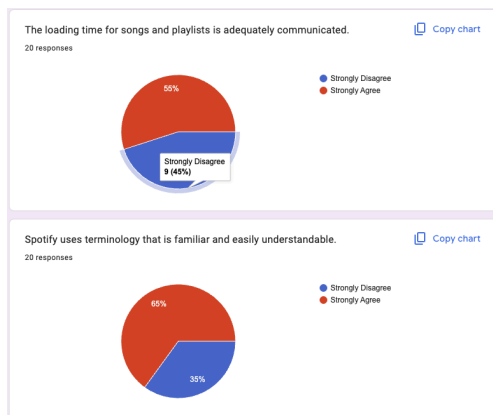
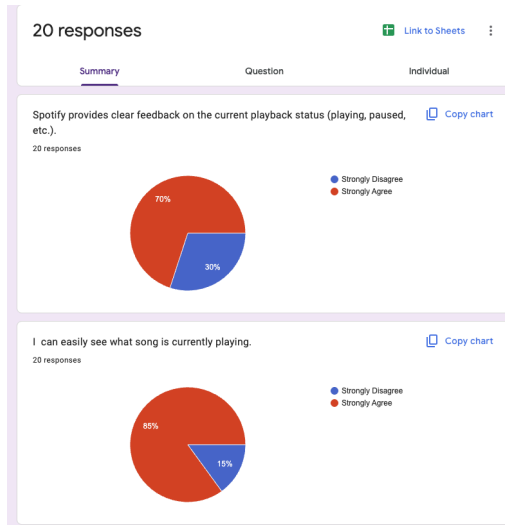
The following questions were included in the Spotify Listening Insights Needfinding survey, it had 25 participants and it was done through Google Forms.

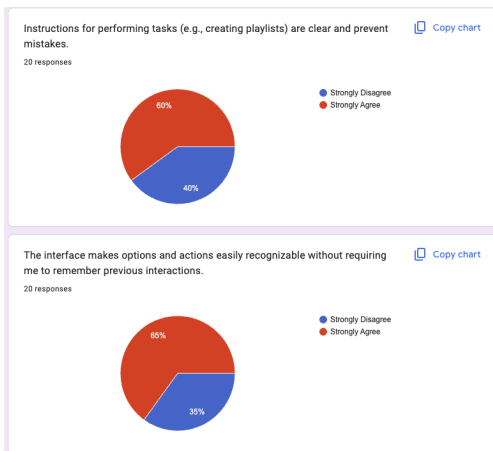
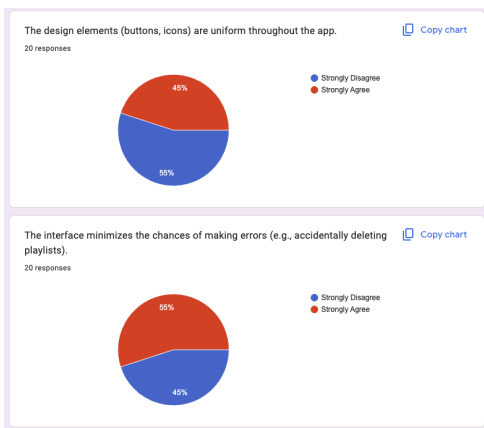
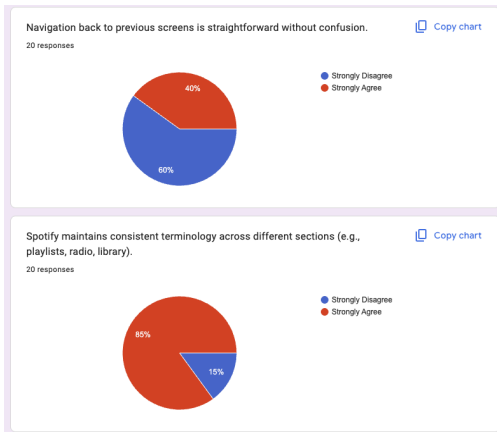
- How often do you use Spotify? (Daily/ Weekly/ Monthly/ Rarely)
- How long have you been using Spotify? (Less than 6 months/ 6 months – 1 year/ 1–3 years/ More than 3 years)
- What type of Spotify account do you have? (Free, Premium)
- What is your age range? (Under 18/ 18-24/ 25-34/ 35-44/ 45+)
- How do you typically listen to Spotify? (Mobile app/ Desktop app/ Web browser)

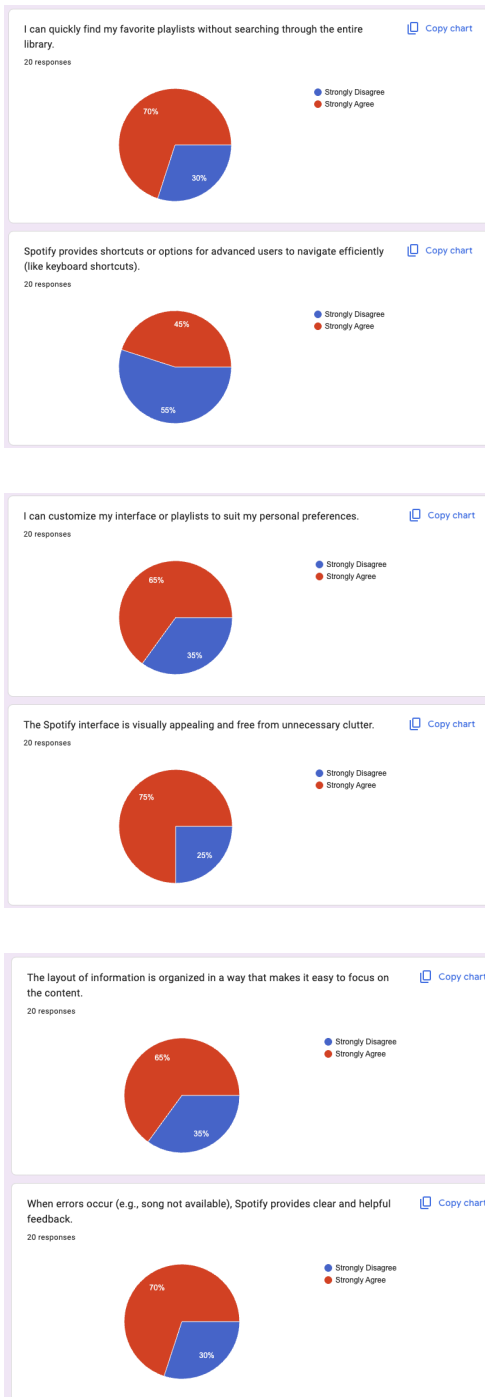
- What genres do you listen to the most? (Select all that apply: Pop/ Rock/ Hip-Hop/ Classical / etc.)
- What devices do you primarily use Spotify on? (Select all that apply) (Smartphone, Laptop/Desktop, Tablet, Smart speaker, Car system)
- How satisfied are you with the current Spotify interface? (1-5 scale, with 1 being very dissatisfied and 5 being very satisfied)
- Please rate your satisfaction with the following Spotify features (1-5 scale): (Overall Interface/ Library Organization/ Search Functionality/ Recommendation Engine/ Yearly Wrap Feature)
- How frequently do you use the following features? (Never, Rarely, Sometimes, Often, Always)
 - Playlists
 - Discover Weekly
 - Daily Mixes
 - Podcasts
 - Collaborative Playlists
- How familiar are you with Spotify Wrapped? (Scale: 1 = Not familiar, 5 = Very familiar)
- How much do you enjoy Spotify Wrapped each year? (Scale: 1 = Not enjoyable, 5 = Extremely enjoyable)
- What do you usually do with your Spotify Wrapped?
 - Just view it
 - Share on social media
 - Compare with friends
 - I usually skip it
- How interested are you in seeing detailed statistics about your music listening habits? (Scale: 1 = Not interested, 5 = Very interested)
- How often would you like to see summaries of your listening data?
 - Weekly
 - Monthly
 - Quarterly
 - Only once per year
- Which types of listening insights would interest you the most? (Select up to 3)
 - Top songs
 - Top artists
 - Genre distribution
 - Listening time (hours per week/month)
 - Music trends over time
 - Mood or listening patterns

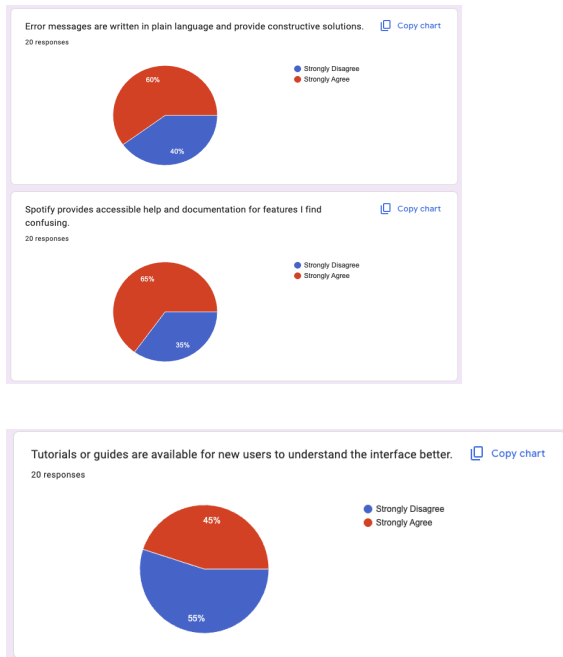
- Music discovery statistics
- Have you ever used third-party apps or websites to see your Spotify listening statistics? (Yes/No)
If yes, which ones? (Open text)
 Why did you use those tools instead of Spotify's built-in features? (Open text)
- How easy is it currently to find insights about your listening habits within Spotify? (Scale: 1 = Very difficult, 5 = Very easy)
- What frustrates you most about Spotify's current statistics features? (Open text)
- If Spotify had a dedicated "Listening Insights" dashboard, how likely would you be to use it? (Scale: 1 = Very unlikely, 5 = Very likely)
- What additional information or insights would you like Spotify to show about your music listening habits? (Open text)
- How do you prefer to view your data? (Select all that apply)
 - Graphs (e.g., bar, line)
 - Pie charts
 - Textual summaries
 - Interactive dashboards
- Would you be interested in a feature that provides insights like:
 - Time spent listening per genre?
 - Comparison of listening habits over several months?
 - Visual representations of your listening patterns? (Yes/No, with space for elaboration)
- What specific features do you think would enhance your experience with Spotify? (Open-ended)

Appendix B Heuristic Evaluation Survey









Appendix C AI BRAINSTORMING

help me brainstorm for this theme projrt, ideas: Redesigning Spotify's
Listening Insights: Improving Access to Personal Music Statistics
Beyond Spotify Wrapped

Great theme—you already have a strong foundation with Spotify and its popular feature Spotify Wrapped. The key opportunity is making insights *continuous*, *actionable*, and *explorable* instead of once-a-year and static.

Here are **20 concise brainstorming ideas** you can use for your project:

Access & Visibility Improvements

1. Add a persistent "Insights" tab on the main navigation for always-available stats.
2. Introduce a dashboard homepage with real-time listening metrics (daily, weekly, monthly).
3. Allow users to pin favorite stats (e.g., top genres) to their home screen.
4. Create a searchable insights feature (e.g., "When did I last listen to Drake?").
5. Add timeline sliders to explore listening history across time periods.

Data Visualization Enhancements

6. Use interactive graphs showing listening trends over time (genres, moods, artists).
7. Build heatmaps of listening activity by time of day or day of week.
8. Show "music evolution journeys" (how taste changes over months/years).
9. Compare listening stats across different periods (this month vs last month).
10. Visualize "listening streaks" and milestones (e.g., 7-day genre streak).

Personalization & AI Features

11. Provide AI-generated insights like "You prefer upbeat music on Mondays."
12. Generate weekly mini-Wrapped summaries automatically.
13. Suggest new music based on under-explored listening patterns.
14. Create personality-style summaries (e.g., "You're a nostalgic listener").
15. Let users ask questions about their data (chat-based insights interface).

Social & Sharing Features

16. Enable real-time sharing of stats (not just annual Wrapped stories).
17. Let users compare listening profiles with friends.
18. Add collaborative insights for shared playlists (group listening stats).
19. Create badges or achievements users can share (e.g., "Top 1% listener").
20. Introduce "music identity cards" that update dynamically and are shareable.

HCI / UX Angles You Can Emphasize

- Discoverability (users don't know insights exist outside Wrapped)
- Frequency vs. overload (how often to show insights without annoying users)
- Interpretability (making stats meaningful, not just numbers)
- Engagement design (turning passive data into interactive experiences)

Appendix D SURVEY RAW DATA — NEEDFINDING

Summary statistics and response distribution for all 25 survey responses are presented below.

Survey Item	Percentage	Count (of 25)
Spotify usage frequency (Daily)	44%	11 of 25
Account type (Premium)	52%	13 of 25
Age range 25–34	68%	17 of 25
Overall interface satisfaction (4–5)	64%	16 of 25
High interest in listening stats (4–5)	68%	17 of 25
Would use Insights dashboard (4–5 likely)	44%	11 of 25
Prefer graphs for data display	68%	17 of 25
Interested in genre/time/visual features	84%	21 of 25
Used third-party stats apps (Yes)	8%	2 of 25

Table A1. Key quantitative results from the 25-participants needfinding survey.